

pedestrian interactions than autonomous cars.” Due to this, there are a few areas where autonomous delivery robots are acting as a precursor to autonomous cars.

First, he explains, the nuances of running an autonomous fleet of ten vehicles compared to a fleet of 200 to 2,000 delivery robots is a huge shift in AV operations. The educational benefits of autonomous robot fleets include, and are not limited to, autonomous fleet management, distributed AV information and processing, fallback safety, fleet orchestration, and more. Second, because more delivery robots are out in the market, they are more exposed to a variety of edge cases. Sometimes, sidewalks, airports, and curb sides are more challenging than lane-guided roads for autonomous cars, and that results in further advancements in active perception, situational awareness, and autonomous behaviours and safety (in general).

Design, of course, is a different ballgame. The purpose of delivery gives rise to certain special requirements. Obviously, storage is an important aspect of the design. This comes in various forms—modular cargo boxes, simple locker boxes, and so on—depending on the size and purpose of the robot. Some have hot and cold compartments as well. They also need to look friendly and have a safe, harmless design with soft, rounded corners. Some like Nuro even have airbags in the front to protect pedestrians from harm in case of a bump.

Delivery robots have to function in line with local rules and regulations at all times. They can move only where they are permitted. And features like lights and alarms have to be built in, so that fellow pedestrians get a clear indication when and where the robot is moving. Delivery robots also have to be very courteous and nice to people. They are taught to say, *excuse me*, when someone is blocking the way, and *thank you*, after they oblige and move!

“It’s a never-ending journey to make sure that this human-robot dynamic



MK27-2, Amazon Prime Air’s latest delivery drone, with a unique hexagonal design for better stability and six degrees of freedom (Courtesy: Amazon)

is maintained, fostered, and improved. When the robot makes an error or does something unexpected, we try to take all of these points of feedback in to make sure that it gets better and better, because the bar gets raised as well. We can’t assume the bar is static. I am of the opinion that if you’re going to automate something, it not only needs to be as good but it needs to be better than a human being,” says Westgarth.

It is never man vs machine

Although many look at autonomous delivery as a replacement for human workers, as something that’s going to steal their jobs, that might not always be the case. It relieves them from monotonous jobs, gruelling night shifts, extended work hours, and hazardous conditions, freeing human labour for more subjective and valuable tasks.

Westgarth of Starship Technologies said in a recent interview, “We believe that if you can insert value into the economy overall, whatever jobs the economy provides will be of higher value. One of the interesting dynamics that we saw in some environments was

that our partners and customers would have a hard time hiring staff. However, when we posted job openings for staff—technical staff, project management staff or whatever—we would have no trouble hiring them. They would come to us. It is because they saw more opportunity to be involved in an evolving economy. To them it was more attractive working with robotics than making food or delivering food themselves. Both are extremely valuable. But at the end of the day, if we could raise the value of what the human being does versus what this automated machine does, I think that’s a good change in society overall.”

Westgarth cites the example of automation in the automotive industry. The level of automation today is exponentially higher than it was 20-30 years earlier. Superficially, we would think that all this automation would have taken so many jobs away from human workers. However, there are many more workers in the automotive industry today than before. So, the situation is quite nuanced, and not as easy as saying automation will take away the jobs.

Honestly, you cannot stop technology, not at this stage of growth; so it is better to find a balance. Unlike self-driving cars and heavy vehicles, autonomous last-mile delivery systems, especially mobile robots, seem well-poised for growth in the near term. Trendwatchers believe that more countries will start giving permissions for small and slow AVs to ply on sidewalks, roads, and public places, unlike the hesitance with regards to fast cars. Slowly, the laws of the economy will also play their magic, and find a balance between human-powered and machine-powered tasks. **END** 🐧

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